

**Ubuntu Innovations
(Pty) Ltd**



IT Operating Systems & User Access Report

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Table of Contents

Executive Summary & Introduction	2
Access Control Policy and Security Group Mapping	2
Who gets which key	2
Administrative Escalation and Auditing Oversight.....	3
Network-Attached Storage (NAS) Shared Folder Architecture	4
Governance and Objectives.....	4
Architectural Design Principles.....	4
Root Directory Topology	5
<i>Data Repository Taxonomy ("What Lives Where")</i>	6
Identity and Access Management (IAM) Protocol	7
File System Taxonomy and Naming Conventions	8
Operational Governance and Data Stewardship.....	9
Backup Infrastructure and Retention Policy	9
Governance Review and Change Management Schedule.....	9
Command-Line Administration	10
Systems Administration and Directory Security.....	10
Terminal Log Output and Directory Permission Verification	11
Software Deployment Plan.....	12
Endpoint Management and Deployment Strategy	12
IP Addressing Scheme and Subnetting.....	13
Network Segmentation and Access Control.....	13
Conclusion & Outlook.....	14

Executive Summary & Introduction

As **Ubuntu Innovations (Pty) Ltd.** continues to expand its operational footprint, the requirement for robust, reliable, and secure information technology infrastructure becomes paramount to maintaining operational efficiency and data integrity. This technical report outlines the strategic implementation of the company's systems administration baseline, endpoint deployment frameworks, and centralized storage architectures.

The objective of this initiative is twofold: first, to establish rigorous role-based access controls and secure data directory management via localized systems administration; and second, to modernize our enterprise data storage through a structured Network-Attached Storage (NAS) framework. By transitioning **Ubuntu Innovations (Pty) Ltd.** from legacy, flat-network environments to optimized, segmented subnets, this deployment ensures absolute data confidentiality, simplifies firewall administration, and provides a highly scalable foundation capable of supporting ongoing corporate growth.

Access Control Policy and Security Group Mapping

To maintain a secure and scalable network ecosystem, system access is governed strictly through an enterprise **User Permissions Matrix**. Rather than applying explicit security permissions to individual user accounts—an administrative practice that introduces auditing complexures during employee turnover—permissions are mapped exclusively to functional security groups.

Utilizing this structural framework, each directory operates as a secured boundary, where membership within a corresponding Active Directory (AD) security group acts as the cryptographic credential required for access. Users who are not explicitly enrolled in the designated group are systematically denied entry, ensuring absolute data isolation across the enterprise.

Who gets which key

Team	Security group	What they can get into
Management	mgmt_grp	Full access across the board. They can open and change anything.

Finance	finance_grp	Finance folder only. Everything else is closed off.
HR	hr_grp	HR folder only. Same deal — nothing outside their area.
Sales	sales_grp	Sales folder only.
Development	dev_grp	Dev folder only.

Administrative Escalation and Auditing Oversight

While executive management retains comprehensive administrative access privileges across all file system directories, day-to-day operational data management remains delegated strictly to respective department personnel. Executive-level access is maintained primarily to facilitate independent compliance audits, ensure administrative redundancy, and allow for formal directorial verification or sign-off on isolated assets during exceptional operational scenarios.

Network-Attached Storage (NAS) Shared Folder Architecture

Governance and Objectives

The purpose of this document is to establish a standardized logical architecture for the organization's central Network-Attached Storage (NAS) system. Unstructured data growth and ad-hoc folder creation inevitably led to operational inefficiencies, data fragmentation, and security vulnerabilities. By defining a rigid directory hierarchy and formal data governance policies from the outset, the organization ensures long-term data integrity, ease of navigation, and auditable access control.

Architectural Design Principles

The storage framework is governed by five core operational principles:

- **Intuitive Taxonomy:** Folder nomenclature relies on clear, unambiguous, and standardized terminology to eliminate operational confusion and misfiling.
- **Principle of Least Privilege (PoLP):** User access is strictly restricted to the minimal data required to fulfill specific job functions. This limits internal data exposure and prevents accidental modification or deletion of critical assets.
- **Standardized Cross-Departmental Templates:** A uniform directory structure is deployed across all business units, fostering administrative consistency and reducing user learning curves.
- **Scalable Architecture:** The directory framework is designed to accommodate organizational expansion, allowing new teams or business units to integrate seamlessly without disrupting the existing hierarchy.
- **Strict Data Isolation:** High-risk departments, specifically Finance and Human Resources, are logically isolated within the directory tree to guarantee strict confidentiality and regulatory compliance.
- **Automated Disaster Recovery:** The entire /Company root directory tree is integrated into an automated nightly backup schedule to ensure business continuity.

Root Directory Topology

To maintain a clean and manageable file system, all organizational data resides beneath a single root directory designated as /Company. The secondary tier of this hierarchy is strictly categorized by department or specific cross-functional business purposes.

/Company

├── Public/ (Global Access – Corporate Distribution)

| ├── Announcements/

| ├── Templates/

| └── Policies/

├── Finance/ (Restricted – Financial Operations)

| ├── Invoices/

| ├── Payroll/

| ├── Budgets/

| └── Reports/

├── HR/ (Restricted – Human Resources)

| ├── Employee-Records/

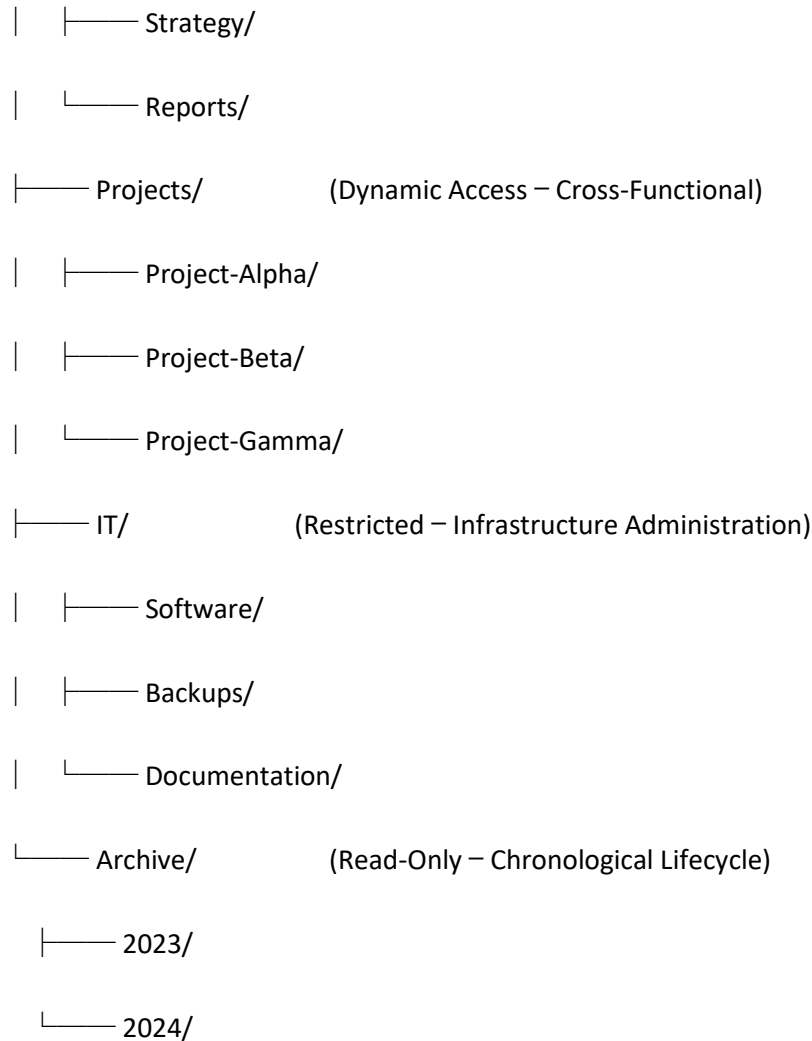
| ├── Recruitment/

| ├── Contracts/

| └── Training/

├── Management/ (Restricted – Executive Leadership)

| ├── Board-Meetings/



Data Repository Taxonomy ("What Lives Where")

To maintain operational security and structural integrity, data is categorized into distinct, functional repositories with tailored access control lists (ACLs):

- **/Company/Public (General Distribution):** This directory serves as the centralized repository for organizational announcements, document templates, and the corporate handbook. It is configured with global write privileges to allow seamless collaboration on non-sensitive corporate assets.

- **/Company/Finance (Financial Operations):** Reserved exclusively for invoices, payroll processing, budgetary frameworks, and fiscal reporting. Access to this directory is strictly audited, logged, and restricted to personnel with authorized financial clearance.
- **/Company/HR (Human Resources):** Contains confidential employee records, employment contracts, and recruitment pipelines. Access is tightly controlled and restricted solely to HR personnel to maintain privacy compliance.
- **/Company/Management (Executive Strategy):** Houses executive board papers, strategic planning documentation, and director-level reporting assets. Access is explicitly restricted to senior management and executive directors.
- **/Company/Projects (Cross-Functional Operations):** Organized dynamically using dedicated subfolders for each active initiative. Because project teams draw members from multiple business units, access permissions are provisioned on a per-project basis rather than by department.
- **/Company/IT (Information Technology):** Reserved for software installer images, systems infrastructure documentation, network configuration mapping, and backup logs. Access is limited strictly to IT administration personnel.
- **/Company/Archive (Data Lifecycle Management):** Formats historical data and completed project files as read-only assets, structured chronologically by calendar year. To preserve data integrity and prevent unauthorized extraction, no assets may be moved out of this directory without formal executive sign-off.

Identity and Access Management (IAM) Protocol

To ensure scalable and maintainable security administration, our team enforces **Role-Based Access Control (RBAC)**.

System permissions are mapped strictly to functional security groups rather than individual user accounts. This architectural standard streamlines employee onboarding and offboarding procedures; administrators only need to modify group memberships rather than auditing or reconfiguring permissions across disparate file systems.

Folder	Read	Write	Full control
/Company/Public	All Staff	All Staff	IT Admins
/Company/Finance	Finance Team	Finance Team	Finance Manager, IT
/Company/HR	HR Team	HR Team	HR Manager, IT
/Company/Management	Managers	Managers	Directors, IT
/Company/Projects	Project Members	Project Members	Project Lead, IT
/Company/IT	IT Team	IT Team	IT Admins
/Company/Archive	Dept Managers	(read-only)	IT Admins

File System Taxonomy and Naming Conventions

To ensure long-term data discoverability and maintain automated scripting compatibility, the organization adheres to a strict file and directory nomenclature standard:

- **Explicit Nomenclature:** Directory and file names must be fully articulated. Abbreviations are prohibited to eliminate ambiguity (e.g., utilizing Employee-Records rather than EmpRec).
- **Delimiter Standardization:** Spaces within file and directory names are prohibited, as they introduce compilation and syntax errors during automated script execution. All spaces must be replaced with hyphens (-).
- **Chronological Ordering Syntax:** All date-dependent files must utilize the ISO 8601 standard (YYYY-MM-DD) as a prefix to ensure optimal linear sorting within file explorers (e.g., 2026-05-27-Invoice-Acme.pdf).
- **Project Directory Formatting:** Active project folders must consistently follow the structural syntax prefix of Project-<Name>.
- **Archival Subdirectory Structure:** Subfolders within the archival directory are to be labeled strictly by the four-digit calendar year (YYYY).

Operational Governance and Data Stewardship

Clearly defined operational roles ensure accountability and protect system resources across three organizational tiers:

- **Information Technology (IT) Department:** Responsible for overall NAS infrastructure management, Active Directory security group maintenance, system updates, and backup integrity.
- **Departmental Managers:** Act as primary data owners. They hold the administrative authority to approve, audit, and modify user memberships within their respective department-level security groups.
- **General End-Users:** Responsible for day-to-day data compliance. Users must ensure assets are saved within their designated directory structures and strictly refrain from placing confidential or sensitive data within the globally accessible Public directory.

Backup Infrastructure and Retention Policy

To mitigate data loss risks and fulfil regulatory compliance requirements, the organization enforces a multi-tiered backup and retention framework:

- **Off-Site Disaster Recovery:** The entire /Company root directory tree undergoes an automated backup sequence to a secure, off-site storage repository on a nightly baseline.
- **Short-Term Snapshot Recovery:** Point-in-time system snapshots are retained for 30 days to facilitate rapid restoration of accidentally modified or deleted files.
- **Long-Term Data Retention:** In accordance with statutory requirements, the /Company/Archive tree is preserved in a permanent, immutable state for a minimum retention period of seven (7) years.

Governance Review and Change Management Schedule

This storage architecture and governance policy is a living document. The IT department, in coordination with departmental business leaders, will conduct a formal structural audit every six (6) months. Out-of-band reviews will be triggered immediately in the event of significant

organizational restructuring, mergers, acquisitions, or any changes that alter the underlying operational assumptions of this framework.

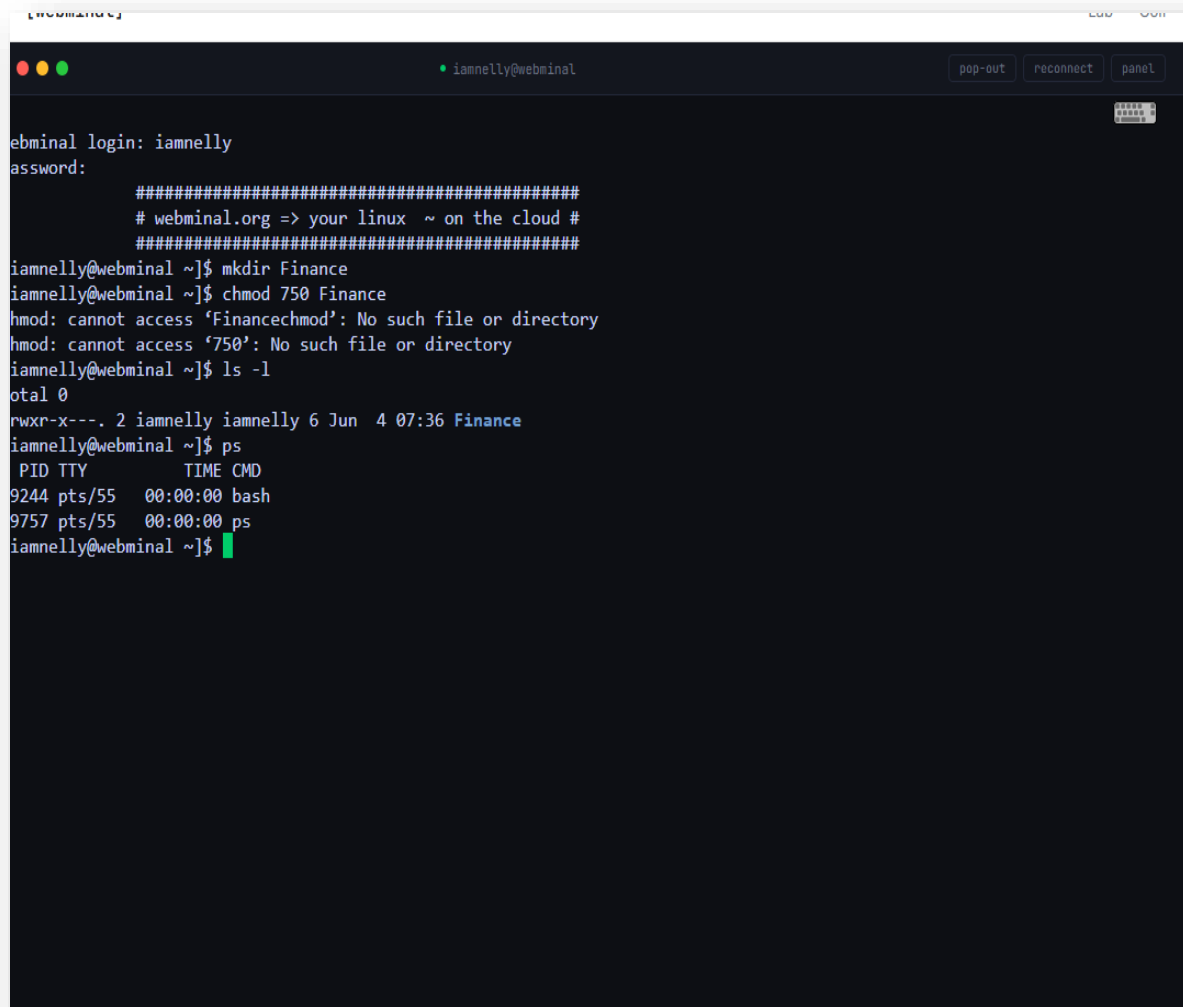
Command-Line Administration

Systems Administration and Directory Security

As part of our infrastructure deployment strategy, our team utilized the Linux command-line interface to establish secure directory structures, manage access control permissions, and audit active system processes.

- **Directory Initialization:** First, we executed the `mkdir Finance` command to provision a dedicated, isolated directory for financial data within the corporate file system.
- **Access Control and Security Compliance:** Next, we hardened the directory's security posture by applying a `chmod 750 Finance` configuration. This specific permission scheme enforces strict role-based access control: it grants the administrator full read, write, and execute privileges (7), allows authorized group members to view and execute resources (5), and completely restricts unauthorized users (0) from accessing sensitive financial assets.
- **System State Verification:** To verify that our security policies were accurately applied, we utilized the `ls -l` command to generate a comprehensive directory listing. As demonstrated in our system logs below, the resulting `drwxr-x---` permission string confirms that the 750 access controls are active, correct, and compliant with data isolation standards.
- **Active Process Auditing:** Finally, we ran the `ps` command to monitor system activity within the current operational session. The generated report verified that only the core command interpreter (`bash`) and the active diagnostic process (`ps`) were running, confirming an optimal and secure execution environment.

Terminal Log Output and Directory Permission Verification



The screenshot shows a terminal window titled "webminal" with a user "iamnelly@webminal". The terminal output is as follows:

```
webminal login: iamnelly
password:
#####
# webminal.org => your linux ~ on the cloud #
#####
iamnelly@webminal ~]$ mkdir Finance
iamnelly@webminal ~]$ chmod 750 Finance
chmod: cannot access 'Financechmod': No such file or directory
chmod: cannot access '750': No such file or directory
iamnelly@webminal ~]$ ls -l
total 0
-rwxr-x---. 2 iamnelly iamnelly 6 Jun  4 07:36 Finance
iamnelly@webminal ~]$ ps
  PID TTY          TIME CMD
 9244 pts/55    00:00:00 bash
 9757 pts/55    00:00:00 ps
iamnelly@webminal ~]$
```

On the right side of the terminal window, there is a sidebar with the following elements:

- A header "IA iamnelly connected" with a green status indicator.
- A button labeled "Lessons".
- A section titled "// SELECT LESSON" with a "Select" button.
- A message: "Select a lesson" and "Content will appear".
- A "prev" button at the bottom.

Software Deployment Plan

Endpoint Management and Deployment Strategy

To maximize operational efficiency for our fleet of 25 workstations, our team will utilize centralized deployment rather than manual, machine-by-machine installations. Manual configuration is inefficient and fails to scale alongside organizational growth.

Because an Active Directory (AD) environment is already established, we will leverage native **Group Policy Objects (GPOs)** for centralized software and policy distribution. While enterprise platforms like Microsoft Endpoint Configuration Manager (MECM) or cloud-based Microsoft Intune would be considered for larger or hybrid fleets, GPOs provide a robust, zero-cost solution for an organization of our current scale.

Our deployment framework includes the following baseline initiatives:

- **Standardized Operating System Images:** We will develop a master Windows image pre-configured with essential productivity software (Microsoft Office), endpoint security (antivirus), and standard web browsers. New hardware deployment will be streamlined via Preboot Execution Environment (PXE) network booting or standardized USB cloning.
- **Application Repackaging:** Software installers that do not natively support standard installation formats will be repackaged into .msi or .msix formats. This ensures all deployments execute silently in the background without user intervention.
- **Automated Patch Management:** Software updates and system pushes will be scheduled outside of standard business hours. Enforcing automated update schedules prevents configuration drift and ensures consistent security compliance across the network.

Ultimately, utilizing existing GPO infrastructure eliminates unnecessary procurement costs and avoids the operational overhead of managing redundant third-party administrative tools.

IP Addressing Scheme and Subnetting

The historical reliance on a flat network architecture introduces significant broadcast traffic risks and scaling bottlenecks as the device count increases. To mitigate this, the network has been segmented into distinct /24 subnets under the private 192.168.0.0/16 address space.

Allocating dedicated subnets based strictly on device functionality drastically simplifies firewall policy enforcement, streamlines network administration, and isolates broadcast domains to improve overall network performance.

Subnet	Range	What lives there
Servers & infrastructure	192.168.10.0/24	Gateway, DNS, DHCP, file server, domain controller
User workstations	192.168.20.0/24	All 25 desktops and laptops
Printers & peripherals	192.168.30.0/24	Network printers, scanners, UPS management
Wi-Fi / guest	192.168.40.0/24	Wireless clients and the isolated guest VLAN
Management / IoT	192.168.50.0/24	IP cameras, door access, HVAC controllers

Network Segmentation and Access Control

Implementing network segmentation ensures that broadcast domains are isolated, effectively optimizing traffic flow and reducing overall network noise. Furthermore, this architectural approach simplifies the implementation of firewall policies; rather than managing a complex list of security exceptions, a single global rule can be deployed to restrict guest networks from communicating with production servers.

Additionally, establishing a structured IP addressing scheme enhances troubleshooting efficiency. By assigning distinct IP subnets to specific categories of equipment, administrators can

immediately identify a device's functional role based solely on its network address, significantly reducing the time required for system diagnostics.

Conclusion & Outlook

The structural configurations detailed within this report establish a verified, secure, and highly manageable operational baseline for **Ubuntu Innovations (Pty) Ltd.**'s current fleet of 25 workstations and centralized data repositories. By leveraging native Active Directory environments and Group Policy Objects, the organization achieves enterprise-grade endpoint configuration management and patch automation without incurring unnecessary software procurement costs. Simultaneously, the newly deployed Network-Attached Storage (NAS) architecture enforces strict data isolation for high-risk assets, standardizes file naming conventions, and ensures automated disaster recovery resilience.

Moving forward, the systematic six-month governance review schedule will ensure that these access control frameworks, IP addressing subnets, and storage taxonomies adapt dynamically alongside evolving business operations. Ultimately, this infrastructure deployment successfully mitigates security risks, eliminates administrative bottlenecks, and guarantees structural data compliance across the entire enterprise of **Ubuntu Innovations (Pty) Ltd.**